

FIT9132 - Database Assignment 1

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TASK 2 - Ocean Odyssey Normalization

Cruise ID 1234
Cruise Name Sydney Brisbane Getaway
Cost per person \$810

Port Code	Port Name	Port Latitude	Port Longitude	Country code	Country Name	Departure/Arrival	Date/Time
AUSYD	Sydney	-33.865143	151.209900	AUS	Australia	Depart	19 Feb 2025 15:30
AUBNE	Brisbane	-27.470125	153.021072	AUS	Australia	Arrive	21 Feb 2025 07:00
AUBNE	Brisbane	-27.470125	153.021072	AUS	Australia	Depart	21 Feb 2025 17:00
AUSYD	Sydney	-33.865143	151.209900	AUS	Australia	Arrive	23 Feb 2025 08:00

UNF:

CRUISE (cruise_id, cruise_name, cost_per_person, (port_code, port_name, port_latitude, port_longitude, country_code, country_name, departure_arrival, date_time))

1NF:

CRUISE (cruise_id, cruise_name, cost_per_person)

CK:

cruise_id

ITINERARY (cruise_id, ~~port_code~~, date_time, port_name, port_latitude, port_longitude, country_code, country_name, departure_arrival)

CK:

cruise_id, port_code, date_time

Partial Dependencies:

~~port_code~~ -> port_name, port_latitude, port_longitude, country_code, country_name

2NF:

CRUISE (cruise_id, cruise_name, cost_per_person)

ITINERARY (cruise_id, port_code, date_time, departure_arrival)

PORT (port_code, port_name, port_latitude, port_longitude, country_code, country_name)

Transitive Dependencies:

country_code -> country_name

3NF

CRUISE (cruise_id, cruise_name, cost_per_person)

ITINERARY (cruise_id, port_code, date_time, departure_arrival)

PORT (port_code, port_name, port_latitude, port_longitude, country_code)

COUNTRY (country_code, country_name)

Full Dependencies:

cruise_id -> cruise_name, cost_per_person

cruise_id, port_code, ~~date_time~~ -> departure_arrival

port_code -> port_name, port_latitude, port_longitude, country_code

country_code -> country_name

Final 3NF Relations:

CRUISE (cruise_id, cruise_name, cost_per_person)

ITINERARY (cruise_id, port_code, date_time, departure_arrival)

PORT (port_code, port_name, port_latitude, port_longitude, country_code)

COUNTRY (country_code, country_name)

Attribute Synthesis:

From Task 1, we have the following relations:

OPERATOR (operator_id, company_name, ceo_name)

SHIP (ship_code, ship_name, commission_date, tonnage, max_guest_cap, reg_country)

CRUISE (cruise_id, cruise_name, cruise_desc, depart_date, depart_time)

CABIN (cabin_no, ship_code, sleeping_capacity, cabin_class)

BOOKING (cruise_id, pass_id, cabin_no, boarding_time, boarding_date)

PASSENGER (pass_id, guardian_id, pass_fname, pass_lname, pass_gender, pass_dob, pass_street, pass_town, pass_postcode, pass_country)

From Task 2, we have the following relations:

CRUISE (cruise_id, cruise_name, cost_per_person)

ITINERARY (cruise_id, port_code, date_time, departure_arrival)

PORT (port_code, port_name, port_latitude, port_longitude, country_code)

COUNTRY (country_code, country_name)

We can notice that the relation “CRUISE” happens to be in both Task 1 and Task 2 relations. So we need to merge these two identical relations.

Therefore,

FINAL 3NF Relations:

OPERATOR (operator_id, company_name, ceo_name)

SHIP (ship_code, ship_name, commission_date, tonnage, max_guest_cap, reg_country)

CRUISE (cruise_id, cruise_name, cruise_desc, depart_date, depart_time, cost_per_person)

CABIN (cabin_no, ship_code, sleeping_capacity, cabin_class)

BOOKING (cruise_id, pass_id, cabin_no, boarding_time, boarding_date)

PASSENGER (pass_id, guardian_id, pass_fname, pass_lname, pass_gender, pass_dob, pass_street, pass_town, pass_postcode, pass_country)

ITINERARY (cruise_id, port_code, date_time, departure_arrival)

PORT (port_code, port_name, port_latitude, port_longitude, country_code)

COUNTRY (country_code, country_name)